

Extrusion Cheat Sheet

Theoretical

Based on a modified Ender syringe extruder. Designed for a 3 mm nozzle but easily adapted. A practical starting point for calibrating and printing pastes, gels, inks, clays and similar materials.

0 Before you're able to calibrate the extrusion while printing, you have to **calibrate the E-steps**. These dictate how much the extrusion motor needs to turn to extrude 1 mm of filament.

Start by inputting the theoretical value in the Gcode. Then fine-tune this value by commanding 50 mm extrusion and then measure the actual length.

$$E_{\text{steps}} = \frac{3.0625 * S_m * M * R}{D^2 * p}$$

S_m : motor steps per rev. (typical 200)

M : microsteps (typical 16)

R : gear reduction ratio

D : syringe inner diameter (mm)

p : lead screw pitch (mm/rev.)

Fine-tune

$$E_{\text{new}} = E_{\text{old}} * \frac{\text{commanded } E}{\text{actual } E}$$

Ender 3-V2 printed

70

